Assignment 8

Due: 4/1

Note: Show all your work.

Problem 1 (10 points). Consider the following contingency table.

	C (buys coffee = Yes)	$\overline{C}$ (buys coffee = No)
T (buys tea = Yes)	742	136
$\overline{T}$ (buys tea = No)	66	469

Compute the *lift*, *all-confidence*, *cosine*, *Kulczynski* and *imbalance ratio* measure, and determine whether buying coffee and buying tea are positively correlated, negatively correlated, or not correlated.

Problem 2 (10 points). Suppose you have the following transactional database:

CID	Time	Items
1	1	10,30
1	4	80
1	6	60, 70
2	3	10
2	7	30,40
2	20	60,70,80
3	2	30,50
3	8	70,80
4	2	70
4	10	80
5	1	30, 50
5	5	70

Show all frequent 2-sequences along with their supports. Assume that the minimum support is 40% (or two data-sequences). You must not use any data mining software including R. You must find all frequent 2-sequences yourself.

Problem 3 (10 points). This problem is about mining strong rules from a transactional dataset using R. Use *hw8_p3.csv* data for this problem. It is a slightly modified version of the *groceries.csv* data that is included in R.

(1). Mine all rules with minimum support = 0.5% and minimum confidence = 50% and include the screenshot of the summary in your submission. How many rules are there?
(2). Mine only length 4 rules with minimum support = 0.5% and minimum confidence = 50% and include the screenshot of the summary in your submission. How many rules are there?

(3). Show top 5 rules based on *coverage*.
(4). Show top 5 rules based on *lift*.

Problem 4 (10 points). This problem is about mining frequent sequences and induced temporal rules from a transactional database using R. Use *hw8_p4.csv* data for this problem.

(1). Mine all frequent sequences with minimum support = 40% and include the screenshot of the summary in your submission.
  (1)-1. How many frequent 1-sequences are there?
  (1)-2. How many frequent 2-sequences are there?
  (1)-3. How many frequent 3-sequences are there?
  (1)-4. How many frequent 4-sequences are there?
(2). From the above frequent sequences, mine all induced temporal rules. How many rules are there?
(3). Show top 5 rules based on *confidence*.
(4). Show top 5 rules based on *lift*.

Submission:

Submit all solutions in a single Word or PDF document and upload it to Blackboard. Use *LastName_FirstName_hw8*.docx or *LastName_FirstName_hw8*.pdf as the file name. You must also submit your R code file. Name it *hw8.R*.